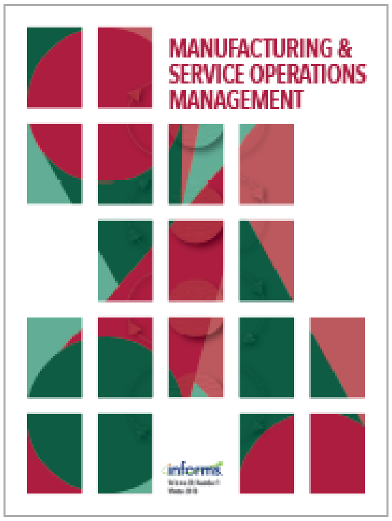


Manufacturing & Service Operations Management

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

Banking on “Mobile Money”: The Implications of Mobile Money Services on the Value Chain

Yan Dong, Moonwon Chung, Chen Zhou, Sriram Venkataraman

informs
http://pubsonline.informs.org/journal/msom/
MANUFACTURING & SERVICE OPERATIONS MANAGEMENT
Vol. 21, No. 2, Spring 2019, pp. 290–307
ISSN 1523-4614 (print), ISSN 1526-5498 (online)

Banking on "Mobile Money": The Implications of Mobile Money Services on the Value Chain

Yan Dong,[a] Moonwon Chung,[a] Chen Zhou,[a] Sriram Venkataraman[a]

[a] Darla Moore School of Business, University of South Carolina, Columbia, South Carolina 29208

Contact: yan.dong@moore.sc.edu, http://orcid.org/0000-0002-3289-4817 (YD); moonwon.chung@grad.moore.sc.edu, http://orcid.org/0000-0002-9579-6026 (MC); chen.zhou@moore.sc.edu (CZ); sriram.venkataraman@moore.sc.edu (SV)

Received: March 31, 2016
Revised: February 10, 2017; October 17, 2017; February 24, 2018
Accepted: March 6, 2018
Published Online in Articles in Advance: August 9, 2018

https://doi.org/10.1287/msom.2018.0717

Abstract. ***Problem definition***: This study examines the effects of mobile money on the value chain, that is, mobile network operators (MNOs), banks, and end users. Mobile money is an innovative technology that is bundled with related services and is designed to provide cost-efficient financial inclusion for underserved populations in the developing world. ***Academic/practical relevance***: By studying the interaction between mobile money value chain structures and service bundle compositions, we expand our understanding on value chain revenue allocation mechanisms in developing economies. This study provides a valuable foundation for practitioners to better coordinate the value chain and facilitate sustainable growth of mobile money services. ***Methodology***: By combining proprietary and public data, we assess the impact of launching mobile money and the potential asymmetric effects of credit payments on the performance of the value chain. We identify control groups by using propensity score matching and estimate a quasi-experimental difference-in-difference regression. ***Results***: We find that the launch of mobile money has had a positive effect on the performance of the value chain, consisting of old, poor, and undereducated populations. However, although some bundled, complementary services, such as credit payments, benefit participating banks, they may be associated with lower MNO profits. Moreover, the effects of mobile money on the value chain remain positive and stable over time, whereas the effects of credit payments are stronger over time. ***Managerial Implications***: Findings suggest that MNOs and banks should be encouraged to launch mobile money and expand the customer base to old, poor, and less-educated members of the population. Certain services such as credit payment may yield asymmetric benefits such that banks could consider compensation mechanisms for the MNOs. Policy makers in developing countries should encourage coordination and promote a fair distribution of benefits between the two partners to sustain the growth of mobile money services.

History: This paper has been accepted for the *Manufacturing & Service Operations Management* Special Issue on Value Chain Innovations in Developing Economies.
Supplemental Material: The online appendices are available at https://doi.org/10.1287/msom.2018.0717.

Keywords: mobile money services • mobile technology innovation • developing economies • empirical research

1. Introduction

Mobile money is an innovative mobile technology that "allows users to deposit money onto their telephone handsets, transfer e-money to another user with a simple text message, and withdraw cash at one of thousands of outlets throughout the country" (Mbiti and Weil 2013, p. 369). Mobile money reduces the financial transaction costs of users (Jack and Suri 2014), is less dependent on physical outlets (e.g., bank branches), and has service features that are easy and straightforward to understand (Lal and Sachdev 2015). Therefore, mobile money combines an innovative business model, value-added services, and digital currency technology for the underserved population (i.e., people without bank accounts) in the developing world (McGrath and Lonie 2013). The value-added services enabled by mobile money include credit payments, salary disbursements, microinsurance, utility bill payments, and international transfers. These services may be complementary and may benefit the value chain of mobile money, which consists of mobile network operators (MNOs), participating banks, and end users. In this sense, mobile money is an attempt to achieve both dimensions of "doing good" and "doing well" through value chain innovation (Lee and Tang 2017). However, bundling services across value chain members is complicated, and the benefits of bundling may be misaligned between the members (Bhargava 2016). In this research, we study the impacts of mobile money on value chain performances and how such impacts are influenced by the bundling of mobile money and a value-added service, that is, credit payments.

From the earliest services like Safaricom's M-PESA, MNOs attempted to leverage their mobile money platform with various value-added services to enhance the end-user experience (Hughes and Lonie 2007). Evidence indicates that these value-added services provide early success in the value chain (Jack and Suri 2014, Mbiti and Weil 2013). For example, Telesom ZAAD in Somaliland integrated mobile money with airtime purchasing services; in the subsequent year, its churn was halved, from 5% to 2.5%, and its airtime sales increased by 33% (Lal and Sachdev 2015). When such services are bundled with complementary banking services, they can generate new revenue streams by allowing the MNOs and banks to solidify their relationships with end users (Lal and Sachdev 2015). For example, GCASH in the Philippines incorporated bulk payment services that allowed business customers to make salary disbursements through mobile money accounts, which saved GCASH administrative costs and operational risks from dealing with large numbers of in-person cash payments for business organizations (Lal and Sachdev 2015).

Among the various value-added services, a credit payment service is particularly important for end users because it allows them to request short-term loans via mobile money, purchase products from merchants, cope with emergencies, and fund education (Cook and McKay 2015). On the supply side, credit payments involve loans as a core business for financial institutions (Sealey and Lindley 1977) and may benefit the service providers with increased customer base and revenues. However, credit payments are subject to regulatory requirements for financial service licensing in most countries (Velayuthan 2010), making it impossible for an MNO to provide this service alone. Therefore, value chain coordination is critical because the two complementary services involve different value chain members, and its sustainability depends on the proper allocation of the benefits between the value chain members. Furthermore, it is important for value chain members to understand how the interdependencies influence the outcomes from services offered by other members. However, extant literature provides limited insights regarding the impact of mobile money and credit payments on each member of the value chain.

Prior research on mobile money has focused on end-user-level issues, such as service quality and inventory of mobile money agent networks, the impact of mobile money innovation on consumption, and characteristics of household transaction networks (Jack et al. 2013, Lal and Sachdev 2015, Mbiti and Weil 2013). Few researchers have examined mobile money in the context of the value chain or the impact on value chain members. In addition, the literature on product bundling shows mixed predictions. Some studies show that bundling strategies can help increase revenues, take advantage of network externalities, and differentiate MNOs and participating banks from their competition (Gawer and Cusumano 2002, Stremersch and Tellis 2002, Prasad et al. 2010). Other studies show bundling across distribution channels may introduce conflicts that may weaken the incentives for bundling (Bhargava 2016), and when all value chains in a market compete by bundling complementary producers, it may lead to overinvestment and lower profits (Mantovani and Ruiz-Aliseda 2016). However, these studies all fall far short in explaining the impact of introducing innovative complementary service bundles in the value chain and the value allocation mechanism between members. Moreover, to the best of our knowledge, the impact of complementary service bundles offered by separate members on the value chain performance has not been studied.

The following important research questions, therefore, arise: How does the innovation of mobile money affect the performance of the value chain? How do bundled mobile money services involving different members in the value chain affect the members' performance? Specifically, we answer the research questions by examining the following: (1) the impact of mobile money on MNOs' and banks' performance, (2) the impact of credit payment value-added service on MNOs' and banks' performance, and (3) the impact of mobile money on end-user welfare in developing markets. We assess these impacts by using proprietary data and by applying quasi-experimental methods.

We find that the launch of mobile money benefits the value chain, that is, MNOs, banks, and end users who are old, poor, and less educated. Our findings make two significant contributions. First, by examining the bundling effects of credit payments and mobile money on value chain performance, we show that whereas the banks gain additional profits from offering complementary services, such as credit payments, the MNOs do not. This suggests an asymmetric distribution of benefits from the complementary services among value chain members. The existing literature focuses on complementary effects of service bundling with one firm, and our findings contribute to the literature by empirically identifying the asymmetric effects of complementarity in the value chain. This further highlights the need for coordination among value chain members regarding value creation and distribution of value from innovative service bundles. Second, we explore the time-varying impacts of mobile money to understand the trajectory leading to sustainable value creation of innovative services. We find that the effects of mobile money on the value chain remain stationary over time, whereas the effects of credit payments are stronger over time. Our findings indicate the necessity of incentive alignment mechanisms for managers and policy makers to ensure sustained success of mobile money and credit payments.

The rest of the paper is organized as follows: Section 2 provides a literature review, and Section 3 describes the research setting, data, and empirical strategy. We present our results in Section 4 and conclude with a discussion and managerial implications in Section 5.

2. Related Literature

In this section, we first review the existing literature on mobile banking and then review the related research on bundling and complementarity, which provides a theoretical lens for our examination of the effect of mobile money and credit payments on the value chain.

2.1. Mobile Banking

Jack and Suri (2014) examine the effects of mobile money innovation on consumption and find that mobile money significantly reduces the transaction costs of money transfers for end users. This reduction in transaction costs should improve risk sharing and should encourage value chain members to participate in the service networks. Focusing on monetary characteristics and cash management at the household level, Mbiti and Weil (2013) describe mobile money as a "gateway via which unbanked households can access financial services" (p. 369) and identify a significant trend of increasing monthly transfer frequency after the launch of M-PESA. This increase in transfers is also confirmed by Jack et al. (2013), who find that mobile money improves efficiency and increases the size of the financial network. Balasubramanian and Drake (2015) study how service quality and competition are associated with demand, electronic credit, and cash; their findings indicate that pricing transparency and agent expertise are both positively associated with demand, whereas competition is positively associated with cash and electronic credit.

In summary, extant research on mobile money has focused on end-user-level issues, such as service quality and the inventory of mobile money agent networks, the impact of mobile money innovation on consumption, and the characteristics of household transaction networks (Jack et al. 2013, Lal and Sachdev 2015, Mbiti and Weil 2013). A noticeable gap still exists in understanding the impact of mobile money on MNOs and banks. We fill this important research gap by studying the impact of the launch of mobile money on the performance of MNOs and banks as well as examining mobile money launch's impact on the welfare of end users. When MNOs initiate mobile money, they are often incentivized to implement service transition strategies to increase revenue as a direct source of growth or to increase customer loyalty and reduce churn rates (Fang et al. 2008). As MNOs compete to build networks that connect people, they tend to extract only small profits from millions of low-value transactions (Lonie and Wagener 2012). Mobile money extends the application of the resource base and knowledge possessed by MNOs and helps MNOs exploit synergies between their core service propositions and value-added service bundles. This exploitation of synergies may result in cost savings and competitive differentiation advantages (Markides and Williamson 1994).

2.2. Bundling and Complementarity

Because mobile money is often bundled with other value-added services, such as credit payments, the literature on product, technology, and service bundling creates a relevant theoretical background. Clustering technologies and services allows firms to exploit complementarities from technological and service innovations (Milgrom and Roberts 1990). For example, internal innovation and external knowledge acquisition can be complementary (Cassiman and Veugelers 2016), and when integrating services and their complementary technologies, an MNO or participating bank may achieve greater expected returns through a focused integration strategy (Anderson and Parker 2013). Further, interactions between different upstream activities of a value chain could be substitutes, whereas interactions between upstream and downstream activities of a value chain could be complements (Hess and Rothaermel 2011). In a study on strategic bundling, Stremersch and Tellis (2002) propose that bundling products increase revenue. Furthermore, in highly competitive industries such as telecommunications and banking, a firm can introduce bundles of different products and services (e.g., telebanking or mobile payment services) to better differentiate itself from the competition. Mixed bundling strategies offer both individual products and their bundles, facilitate differentiation, and therefore often dominate other bundling strategies (Stremersch and Tellis 2002). For technological products, such as Internet-related products, the user base is a critical factor in consumer value, and pure bundling—offering only bundled products—is more profitable when marginal costs are low and network externality is high (Prasad et al. 2010). Similarly, Liu et al. (2010) find support for strong complementarities between the demand for core broadband services and other related services like cable television and local phone services. In a distribution channel, conflicts between members may discourage bundling by the downstream member (e.g., a retailer) because incentives for bundling may be misaligned in the channel (Bhargava 2016). In essence, although the bundling of products and services allows firms to take advantage of complementarity, the effects on firm performance can vary depending on bundling strategies and other factors, such as the roles of MNOs and banks in the value chain.

To summarize, although complementarity in the bundling of mobile money with other services may contribute to the profitability and efficiency of the value chain, the distribution of benefits among the value chain members has not been studied, particularly empirically, in the related literature (Lee and Schmidt 2016). On the one hand, the literature on product bundling shows that bundling strategies can help increase revenues, take advantage of network externalities, and differentiate from competition (Gawer and Cusumano 2002, Stremersch and Tellis 2002, Prasad et al. 2010). On the other hand, another literature stream suggests that when bundling involves distribution channels, conflicts in the channels may weaken the incentives for bundling (Bhargava 2016). Also, other studies predict that innovation ecosystems (as defined in Adner 2006) built around complementary producers may be damaging to each value chain member (Mantovani and Ruiz-Aliseda 2016). These two countervailing perspectives, coupled with a research setting involving developing economies, provides a rich perspective to examine the impact of value-added credit payments, bundled with mobile money, on the performance of the participating value chain members.

3. Research Setting, Data, and Empirical Strategy

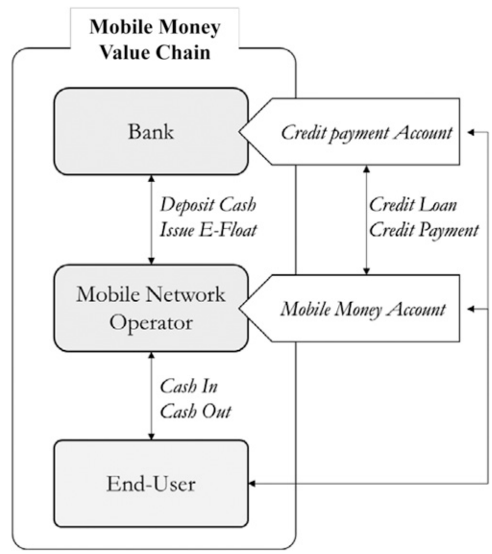
In this section, we first introduce the value chain of mobile money as the background of our research, and then we describe our research setting and data.

3.1. Value Chain of Mobile Money and Credit Payments

We first illustrate the mobile money value chain in Figure 1. Mobile money typically involves incoming, circulating, and outgoing transactions (Almazan and Vonthron 2014). Mobile money allows MNOs to bypass the agents to save the commissions. By bypassing agents, MNOs can sell mobile airtime directly to users and benefit from the mobile money offerings (Leishman 2010). However, these benefits are mostly limited to MNOs. For banks, mobile money may not have a direct value proposition, and it may cannibalize pre-existing bank services. Although MNOs depend on the financial resources from banks to launch and maintain mobile money, incentives are needed for the banks to participate (Gidvani et al. 2016).

Credit payments could be profitable for banks. The payment flow in credit payments is shown in Figure 1. The banks' technological infrastructure and expertise in customer identification and fraud protection, for example, are unique resources that support credit payments

Figure 1. The Structure and Services of the Mobile Money Value Chain



(Cook and McKay 2015, Adengo and Kamukama 2016). However, MNOs' engagement in credit payments reduces the circulating transactions within mobile money accounts and could reduce the MNOs' profits earned from mobile money. Nevertheless, supporting credit payments may be a valid approach for MNOs to encourage active participation of banks in offering mobile money.

3.2. Data

We use multiple proprietary archival data sets for our analysis. Our list of mobile money services, provided by a market research company specializing in the telecommunication industry, spans from 2001 to 2016. This research company operates the mobile money deployment tracker to monitor global mobile money services. The deployment tracker contains information about the name of the service provider and the brand name of the mobile money (e.g., GCASH, M-PESA), the launch date of the mobile money, bundled financial services, and the partners involved in delivering each service. We collected data for 267 launched services. Of these 267 services, 11 were dropped for the following reasons: eight services were launched after 2015, one service lacked a clear launch date, and two merged services had unknown merge dates. Consequently, we assembled a sample of 256 active services and a list of 764 organizations (i.e., MNOs, banks, and third-parties) involved at the time of data collection in January 2016. This number closely represents the entire population of mobile money that had been launched by the time of data collection. We also dropped 38 services provided by third-party organizations. The data contain 184 uniquely identified banks and 183 uniquely identified MNOs in 91 countries, from which we developed three separate panel data sets: one each for MNOs, banks, and end users. Among the remaining 218 worldwide mobile money services in our data set, 77 are operated solely by MNOs, 47 solely by banks, and 94 by MNOs and banks in collaboration. Credit payments are offered by 23 mobile money services, and about half of these (i.e., 11 brands) are jointly operated by both MNOs and banks.

For the MNO data set, we supplemented the mobile money launch data with operational and financial performance metrics of MNOs involved in mobile money. These additional variables were obtained from another proprietary database provided by the market research company, which offers quarterly panel data ranging from the first quarter of 2000 to the fourth quarter of 2014. Using country and MNO name as the key identifiers, we matched a total of 79 countries existing in the two data sets, which left us with 162 MNOs that launched mobile money in their respective countries. We also included MNOs that do not offer mobile money in those 79 countries. In the final sample, we obtained a balanced panel of 407 MNOs across 60 quarters, of which 162 firms launched and 245 firms did not launch mobile money until the fourth quarter of 2014.

For the banks involved, we extracted quarterly performance metrics from the years 2000 through 2014 from Compustat. Because only a few large multinational banks operate across multiple countries, we collected performance metrics of banks in each country where the mobile money was launched. As a result, we obtained an unbalanced panel with 52 banks involved in and 564 banks not involved in mobile money over the 60 quarters.

For the data set on end users, we used Global Findex, an annual survey conducted by the World Bank in 2011 and in 2014. This database is the most comprehensive database on financial inclusion that measures how individuals save, borrow, make payments, and manage financial risks across countries and over time. The data are collected by conducting interviews with about 150,000 randomly selected adults (aged 15 and older) in over 140 countries to construct a representative sample. The database includes both individual-level data (e.g., usage of financial services in the last 12 months) and country-level aggregated data (e.g., percentage of the population using financial services). However, the data set does not provide information on the mobile money used by each surveyed individual. Therefore, we assess the country-level impact of launched mobile money by supplementing our mobile money launch data with the country-level aggregated data from the World Bank. Because we only have two time points of data (2011 and 2014), we restricted our sample to countries in which mobile money was first launched between 2011 and 2014 and to countries in which no mobile money was offered by the end of 2014. Our final sample contains 136 countries that did not have mobile money until 2014 and 25 countries that had mobile money launched between 2011 and 2014 (http://datatopics.worldbank.org/financialinclusion/).

3.2.1. Dependent Variable. Previous studies of the telecommunications industry have used a variety of performance metrics as dependent variables, such as return on assets, return on investment (Bae and Insead 2004), revenue, total voice calls (Majumdar 1998), consumer satisfaction (Gerpott et al. 2001), and churn rates (Ahn et al. 2006). We use a commonly used metric of earnings before interest, taxes, depreciation, and amortization (*EBITDA*) as our focal dependent variable for both MNOs and banks (James 1995, Genakos and Valletti 2011, Kumar and Ravi 2007).

To measure end-user welfare, we focus on three different population groups and assess the impact of mobile money on the percentages of those who had any savings and loans during the past 12 months in each group: the *poor population* (age above 15 in the lower 40% income quantile), the *old population* (age above 65), and the *population with low education* (age above 15 with primary education or less). These measures represent the financial inclusion of the target groups that mobile money is set out to accomplish (Demirgüç-Kunt and Klapper 2012).

3.2.2. Independent Variable. The treatment indicator of the mobile money (*Mobilemoney*) is coded as 1 for all firms that launched mobile money and is coded as 0 for all firms that never launched mobile money within our observed time frame. We use the difference-in-differences analysis to assess the performance changes of firms (i.e., MNOs or banks) that launched mobile money relative to those that did not launch mobile money. We code the binary variable *After* as 1 for all quarters after mobile money is launched for both the treatment group and its constructed control group and 0 for all quarters before mobile money is launched. Our focal independent variable is the interaction term between the *After* and *Mobilemoney* variables. The treatment indicator of credit payments (*Loanlaunch*) is coded as 1 if the mobile money allows end users to make credit payments in circumstances of insufficient currency, and it is otherwise coded as 0. Similarly, we code the binary variable *After* as 1 for quarters after credit payments launch and as 0 for before. We consider the interaction term between the *After* and *Loanlaunch* as the focal independent variable.

To analyze end-user welfare, we use a difference-in-differences analysis to compare the changes in average savings and loans of individuals who lived in countries in which mobile money was launched and those who lived in countries in which mobile money was not available. As explained above, we have restricted our data set to countries that did not have any mobile money launched until 2014 and countries that experienced a launch of mobile money between 2011 and 2014 because of data availability constraints. We code the binary variable *After* as 1 if the year is 2014 and as 0 if the year is 2011. The treatment variable (*Treatment*) is a binary variable that is coded as 1 if the country experienced a mobile money launch between 2011 and 2014. Our focal independent variable is the interaction term between the *After* and *Treatment* variables.

3.2.3. Control Variables. Regarding control variables to be included in the difference-in-differences analyses, we consider country-level characteristics that affect the supply and demand of mobile money. Because we do not observe the total demand in the markets, we include a set of variables that may approximate it (Sharpe 1997). In all the analyses related to MNO performance, we include gross domestic product (*GDP*) and total population (*Population*) to account for the market size and potential of the included countries. On the supply side, we look for exogenous price shifters and potential determinants of marginal cost. As exogenous price shifters, we include the Herfindahl-Hirschman index (*HHI*) of MNOs as a measure of competition intensity. As determinants of the marginal cost of MNOs, we include the real interest rate (*Interest*) (i.e., the average market lending interest rate after adjusting for inflation rates), which serves as a proxy for the price of capital (Evans and Heckman 1984, Grzybowski 2005).

In the analyses involving bank performance, we include all the control variables discussed above except *Interest*. Instead, we use a more detailed firm-level variable of bank net interest margin (*bank net interest margin*) in the matching process, which is described in Section 3.3. Moreover, we use the top-5 bank asset concentration (*Bconcen*) instead of HHI as the measure of competition intensity (Beck et al. 2006, Fries and Taci 2005). In all the analyses related to end-user welfare, we include GDP per capita (*GDPpc*) and total population (*Population*) to control for demographic heterogeneity across countries. We summarize the definitions of our variables in Table 1 and provide the summary statistics of all the variables in Table 2.

3.3. Econometric Models and Estimation Strategy

As discussed earlier, in this research we assess (1) the impact of launching mobile money on the performance of the value chain and (2) the potential asymmetric effects of credit payments on the performance of MNOs and banks if mobile money is launched. The changes in performance after launching mobile money, to a certain extent, indicate the impact of mobile money. However, to make an accurate estimation, we need to benchmark the changes in performance against a control group and apply a difference-in-differences regression. To estimate the impact of launching mobile money, we should compare performance changes of MNOs or banks that launched mobile money to those that never launched mobile money. However, to assess the effects of credit payments, we should compare performance changes of MNOs or banks that launched credit payment service bundled with mobile money to those that launched mobile money only. As a result, we construct different control groups and assess the impact of mobile money and the additional effect of credit payments in separate regressions.

In an ideal scenario, MNOs or banks should be randomly assigned to the treatment group (e.g., launch mobile money) or the control group (e.g., not launch mobile money), and the treatment should start at the same time for all firms in the treatment group. However, in our research context, MNOs or banks chose to launch mobile money, and they chose to do so at different times. In other words, we face two methodological challenges: self-selection of treatment and a sliding window of the occurrence of treatment.

The self-selection of treatment could bias the estimates of the treatment effect (e.g., launching mobile money) if firms decide to launch mobile money based on unobserved factors that relate to performance. For example, MNOs that wished to improve their performance may have decided to launch mobile money, whereas those that had strong performance may have decided not to launch mobile money. In other words, the average performance of the MNOs that chose to launch mobile money may be systematically different from that of MNOs that did not launch mobile money. This difference in average performance may bias the estimated treatment effect if we consider MNOs that launched mobile money as the treatment group and all others as the control group. To resolve this potential issue, we apply the propensity score matching method to construct a control group that is comparable to the treatment group in terms of the likelihood to launch mobile money or credit payments before the occurrence of the treatment (Heckman et al. 1997, Rosenbaum and Rubin 1983). Because we have a relatively small number of treated firms (firms that launched mobile money) compared with the large pool of candidates (firms that never launched mobile money) for the control group, we specify a 5 nearest-neighbor matching. This procedure matches each firm in the treatment group to 5 firms in the control group based on firm characteristics that explain the decision of a firm to launch mobile money or credit payments. The matching ensures that the treatment and control groups are comparable before the treatment occurred. The variables we use to match MNOs are total market share (*Total Marketshare*), total revenue (*Total Revenue*),

Table 1. Variable Definitions of Mobile Money System Characteristics

Model	Variable	Data source	Definition
Focal independent variable	*Mobilemoney*	Mobile money deployment tracker	A dummy variable indicating whether a firm launched mobile money services (treatment group indicator)
	Loanlaunch	Mobile money deployment tracker	A dummy variable indicating whether a firm launched credit payments via mobile money accounts. The service requires credit assessment of the mobile money account holder that is done by financial institutes (treatment group indicator)
Focal dependent variable	*EBITDA*	Proprietary data Compustat	Total earnings before interest, tax, depreciation, and amortization. Collected from financial reports
Consumer dependent variable	*Annual savings*	World Bank	The percentage of respondents (age above 15) who report personally saving or setting aside any money for any reason and using any mode of saving in the past 12 months The variable is measured for three subpopulation groups: poor population (40th income percentile), old population (age above 65), and low education population (primary education or less)
	Annual loans	World Bank	The percentage of respondents (age above 15 for three sub population groups: poor population (40th income percentile), old population (age above 65), and low education population (primary education or less)) who report borrowing any money for any reason and from any source in the past 12 months
Controls	*Bconcen*	World Bank	Assets of five largest banks as a share of total commercial banking assets. Total assets include interest-earning loans, cash, foreclosed real estate, fixed assets, goodwill, other intangibles, current tax assets, deferred tax, discontinued operations, and other assets
	HHI	Proprietary data	Sum squared total of the market share of MNOs within a country
	GDP, GDPpc	World Bank	GDP at purchaser's prices, the sum of gross value added by all resident producers in the economy plus any product taxes and minus any subsidies not included in the value of the products. GDP per capita is calculated by dividing the total GDP by the total population. Data are in current U.S. dollars
	Population	World Bank	The total population is based on the de facto definition of population, which counts all residents regardless of legal status or citizenship
	Interest	World Bank	The lending interest rate adjusted for inflation as measured by the GDP deflator. Calculated as $(I - P)/(1 + P)$, where I is the nominal lending interest rate, and P is the inflation rate as measured by the GDP deflator
Matching covariates	*Total marketshare*	Proprietary data	Total connections at the end of the quarter expressed as a percentage share of the total market connections
	Total connections	Proprietary data	Total unique SIM cards that have been registered on the mobile network at the end of the quarter. A unique subscriber can have multiple connections
	ARPU	Proprietary data	One-quarter lagged average revenue per user, which represents service quality and MNO's capability to extract revenue from each subscriber account
	Total revenue	Proprietary data Compustat	Total income of the MNOs and banks
	Total assets	Compustat	Property, plant, and equipment, which is the long-term or noncurrent asset in the balance sheet. Included are land, buildings, machinery, office equipment, vehicles, furniture, and fixtures used in business
	Total deposits	Compustat	Funding component of a bank from retail, business, and institutional customers
	Bank net interest margin	Compustat	The difference between the interest income generated by banks and the amount of interest paid out to deposits

Table 2. Summary Statistics and Correlation Table of Dependent, Independent, and Control Variables of Interest
MNO variables

Operator variable (n = 879)		Mean	σ	1	2	3	4	5	6	7
1	*EBITDA*	1.19×10^8	1.71×10^8							
2	*Mobilemoney*	0.24	0.43	**0.31**						
3	*Loanlaunch*	0.06	0.24	**0.29**	**0.45**					
4	*After*	0.40	0.49	−0.02	0.05	**−0.08**				
5	*GDP*	5.77×10^{11}	6.30×10^{11}	**0.61**	0.06	**−0.13**	0.01			
6	*Population*	1.76×10^8	3.15×10^8	**0.10**	−0.05	−0.02	0.05	**0.50**		
7	*Interest*	17.44	15.76	**0.25**	**0.16**	0.01	**−0.20**	**0.39**	0.04	
8	*HHI*	3,386.37	1,323.34	**−0.19**	**−0.17**	−0.03	−0.00	**−0.31**	**−0.49**	**−0.53**

Bank variables

Bank variable (n = 5,488)		Mean	σ	1	2	3	4	5	6
1	*EBITDA*	36,689.31	177,115.70						
2	*Mobilemoney*	0.19	0.39	**0.14**					
3	*Loanlaunch*	0.04	0.19	**0.11**	**0.40**				
4	*After*	0.37	0.48	−0.01	0.00	−0.01			
5	*GDP*	4.02×10^{11}	4.99×10^{11}	**0.07**	**−0.09**	−0.02	**0.14**		
6	*Population*	1.49×10^8	2.30×10^8	0.02	**0.07**	0.01	0.01	**0.52**	
7	*Bconcen*	74.81	18.00	**−0.11**	−0.02	**−0.04**	**−0.12**	**−0.40**	**−0.53**

End-user variables

Consumer variable (n = 194)		Mean	σ	1	2	3	4	5	6	7	8	9
1	*Savings (poor 40%)*	35.27	21.81									
2	*Savings (old)*	43.07	22.41	**0.98**								
3	*Savings (low educ)*	15.71	17.84	0.01	0.08							
4	*Loan (poor 40%)*	35.24	15.30	**0.29**	**0.33**	**0.25**						
5	*Loan (old)*	35.37	14.89	**0.23**	**0.30**	**0.37**	**0.95**					
6	*Loan (low educ)*	19.33	19.98	**−0.28**	**−0.23**	**0.74**	**0.53**	**0.61**				
7	*After*	0.49	0.50	**0.33**	**0.33**	**0.22**	**0.18**	**0.18**	0.03			
8	*Treatment*	0.25	0.44	**−0.14**	−0.13	**0.43**	0.08	**0.21**	**0.44**	0.02		
9	*GDPpc*	1.62×10^6	7.11×10^6	0.02	0.05	0.13	0.04	0.03	0.04	0.05	−0.08	
10	*Population*	2.56×10^7	1.04×10^8	0.06	0.10	**0.16**	0.02	0.00	0.04	0.01	−0.04	−0.02

Note. Bold denotes significance at $p < 0.05$ level.

total number of unique subscriber identification module (SIM) cards that have been registered on the mobile network (*Total Connections*), and average revenue per user (*ARPU*). The variables we use to match banks are net interest margin of banks (*Bank Net Interest Margin*), total revenue (*Total Revenue*), total assets (*Total Assets*), and total deposits (*Total Deposits*).

If the treatment occurs to all firms in the treatment group at the same time, we could easily separate the posttreatment periods from the pretreatment periods and compare the average performance changes of the treatment and control groups. The sliding window of the occurrence of treatment complicates the definition of posttreatment periods for the control group. For example, firm A launched mobile money in the second quarter of 2004, whereas firm B launched mobile money in the first quarter of 2006. It is then not straightforward to define which quarters should be the posttreatment periods for firms in the control group. To resolve this issue, we define the posttreatment period of each treated firm as the quarters after it launched mobile money and define the posttreatment period of each control firm the same way as that of the treated firm to which the control firm is matched (Rosenbaum and Rubin 1985). For example, if firm C and firm D are selected to be the counterfactuals of firm A and firm B, respectively, then the posttreatment period of firm C is the second quarter of 2004 and subsequent quarters, and the posttreatment period of firm D is the first quarter of 2006 and subsequent quarters.

3.3.1. Operator Performance. In this section, we describe how we estimate the effect of mobile money on MNO performance and then introduce how we assess the effect of the credit payments on MNO performance. In both cases, we formulate a propensity-score-adjusted difference-in-differences regression (Hirano and Imbens 2001, Robins and Rotnitzky 1995, Rosenbaum and

Rubin 1983, Stuart et al. 2014), which is implemented in two steps. In the first step, we perform propensity score matching to obtain an appropriate control group, along with sampling frequency weights. In the second step, we implement a difference-in-differences regression.

Step 1: Propensity Score Matching. To assess the effect of mobile money, we apply a propensity score matching model to obtain a control group that could serve as a good counterfactual, against which we benchmark the performance change of MNOs after launching mobile money. In this model, we use a Probit regression, where the launch of mobile money is coded as a binary dependent variable. The covariates include factors that potentially influence the decision to launch mobile money. Because a firm could launch mobile money at any period in the 60 quarters of our data, we treat each time period as discrete (Sianesi 2004) and perform the propensity score matching in each given quarter (i.e., 60 times). More specifically, at each quarter t, we use the average values of covariates from quarter *1* to quarter $t-1$ as independent variables in the matching model (Austin 2011). For each MNO i at quarter t, we specify the following Probit model:

$$\Pr(Mobilemoney_{it} = 1 \mid Z_{it}) = \Phi(Z_{it}\beta), \quad \forall t = 1, 2, \ldots, 60$$
$$Z_{it} = \frac{1}{t-1}\sum_{k=1}^{t-1} X_{ik} \tag{1}$$

where X_{ik} is a vector of covariates including *Total Marketshare*, *Total Revenue*, *Total Connections*, and *ARPU*, for firm i at quarter k, and Z_{it} is the vector of these covariates averaged from quarter *1* to quarter $t-1$. An MNO's decision to launch mobile money depends on its financial condition and size of user base because the financial condition determines the feasibility of providing new services (Almazan and Vonthron 2014) and the size of user base affects the expected benefits. The financial condition of an MNO can be captured by *Total Revenue* and *ARPU* of the firm. The size of user base can be captured by the *Total Marketshare* and *Total Connections*.

We use the nearest-neighbor matching algorithm to select five control firms that share the closest propensity scores with each treated firm with replacement. We ensure the quality of the match by checking four statistics, as suggested by Rosenbaum and Rubin (1985) and Rubin (2001): t-tests on the means of covariates of the treatment and control groups, the pseudo R^2 of the Probit regression, the Rubin's B on the absolute standardized difference of the means of the treated and control groups, and the Rubin's R on the ratio of variances of the covariates between the treated and control groups. In addition, we compare the mean and median propensity scores of the treated and control groups and do not find statistically significant differences (at the $p < 0.05$ level). We provide more details regarding the criteria used in matching and the statistics used to assess the quality of matching in Web Appendix WA1. To ensure that the average performance change of the five control MNOs serves as an appropriate counterfactual for that of the corresponding treated MNO, we weigh each treated observation by 1 and each control MNO by sampled frequency divided by 5 (Hirano et al. 2003, Stuart 2010). For example, if a control MNO was assigned to one treated MNO, its weight is one-fifth; if a control MNO was assigned to two treated MNOs, its weight is two-fifths. We then multiply the dependent variable and covariates by the assigned weight (Winship and Radbill 1994, Wooldridge 2010) and use these adjusted values in the difference-in-differences regression.

Step 2: Difference-in-Differences Regression. To assess the effect of launching mobile money on performance of MNO, we formulate a difference-in-differences regression, where the main dependent variable is the profit measure *EBITDA*, and the independent variables are the binary treatment indicator *Mobilemoney*, the binary indicator for the posttreatment periods *After*, and the interaction term of these two indicators. However, the profit of an MNO in a quarter could also be influenced by unobserved firm characteristics, the market condition, and the macroeconomic environment of the given quarter. To control for these factors, we formulate a difference-in-differences panel fixed effects regression with a set of control variables. More specifically, we estimate the following regression

$$EBITDA_{ijt} = \beta_0 + \beta_1 After_t + \beta_2 Mobilemoney_i \cdot After_t + \delta Z_{jt} + \tau_y + \tau_q + \alpha_i + \epsilon_{ijt}, \tag{2}$$

where i indexes MNOs, j indexes countries, and t indexes quarters. The coefficient of the interaction term, β_2 is our key estimator of interest. We drop the main effect of $Mobilemoney_i$ because it cannot be identified when MNO fixed effects are included in the model. Among other variables, the firm fixed effect α_i controls for any unobserved firm characteristics, Z_{jt} is the vector of variables controlling for the market condition (i.e., *HHI*, *Interest*, *Population*, and *GDP*). τ_y and τ_q are sets of year and quarter time fixed effects that control for the macroeconomic environment. We estimate the regression model with standard errors clustered by firms. Our final estimation sample includes 7 treatment MNOs and 24 control MNOs because of missing values.

The identification of the treatment effect depends on the satisfaction of the stable unit treatment value assumptions (SUTVA) (Imbens and Rubin 2015). First, if no treatment occurred, the average profit should be homogeneous between treatment and control groups.

In this research, our propensity score matching method minimizes the differences between each treated MNO and its control MNOs in observed variables that are closely related to profit (i.e., *Total Connections*, *Total Revenue*, and *ARPU*). Therefore, we believe that the treatment and control groups should have homogeneous profit levels if the treatment group did not launch mobile money. Second, the treatment level is consistent across all treated MNOs. Mobile money provides almost identical functions in all markets, such as cash-in/cash-out accounts and text-based transactions. These functions should bring homogenous user experience and benefits, which in turn contributes homogeneously to MNOs' profits. Therefore, we believe that all treated MNOs received a consistent level of treatment.

Step 1: Propensity Score Matching. To assess the marginal effect of launching credit payments in addition to mobile money, we define the treated MNOs as those that launched credit payments as a bundle of mobile money and we obtain the control MNOs from the pool of MNOs that launched mobile money without credit payments. We follow the same matching procedure as the previous subsection to obtain the control group.

Step 2: Difference-in-Differences Regression. We estimate the following difference-in-differences regression:

$$EBITDA_{ijt} = \beta_0 + \beta_1 After_t + \beta_2 Loanlaunch_i \cdot After_t + \delta Z_{jt} + \tau_y + \tau_q + \alpha_i + \epsilon_{ijt}, \quad (3)$$

where all variables are defined identically as Equation (2), except that $Loanlaunch_i$ stands for the binary variable for whether credit payments are included in the bundle of mobile money for firm i, and β_2 captures the key effect of interest. The final sample includes 4 treatment MNOs and 13 control MNOs.

We follow similar arguments as the previous subsection to justify why the SUTVA holds for the identification of the treatment effect. First, the matching covariates (i.e., *Total Marketshare*, *Total Revenue*, *Total Connections*, and *ARPU*) are closely related to profit levels, indicating that the treatment and control groups should have homogeneous profit levels if the treatment group did not launch credit payments. Second, credit payments carry out a homogeneous function of lending short-term loans with high interest fees, which are predominantly spent on airtime purchases or emergency funds across different countries (Cook and McKay 2015). Therefore, the launch of credit payments should homogeneously impact MNOs' profits.

3.3.2. Bank Performance.

Similar processes are implemented to examine bank performance.

Step 1: Propensity Score Matching. A bank's decision to launch mobile money also depends on its financial condition and expected benefits. The financial condition of a bank can be captured by *Total Revenue* and *Bank Net Interest Margin*. The expected benefits can be captured by *Total Deposits* and *Total Assets*. *Total Deposits* are correlated with the customer base of the bank, and it is therefore correlated with the expected benefits from launching mobile money. *Total Assets*, including fixed assets such as property and bank branches, are related to the expected benefits because mobile money allows banks to serve a customer segment without having to establish physical branches, and banks that have fewer physical assets may increase revenue by launching mobile money (DeYoung et al. 2007, Hernando and Nieto 2007). We therefore include *Bank Net Interest Margin*, *Total Revenue*, *Total Assets*, and *Total Deposits* as the matching covariates. After matching, we construct the control group and obtain the weight-adjusted values for the key variables used in the difference-in-differences regression.

Step 2: Difference-in-Differences Regression. As specified in Equation (4), we perform a difference-in-differences regression with panel fixed effects to control for unobserved heterogeneity across banks. Consistent with Equation (2), the main dependent variable for the banks is the *EBITDA*. For bank i in country j at quarter t, we estimate the following equation:

$$EBITDA_{ijt} = \beta_0 + \beta_1 After_t + \beta_2 Mobilemoney_i \cdot After_t + \delta K_{jt} + \tau_y + \tau_q + \alpha_i + \epsilon_{ijt}, \quad (4)$$

where all variables are identical to those in Equation (2), except that (1) the vector of control variables K_{jt} includes *Bconcen* instead of *HHI* to account for competition intensity, (2) K_{jt} does not include country-level *Interest* because *Bank Net Interest Margin*, a matching covariate, captures the variation in interest rates across countries, and (3) α_i represents bank fixed effects. We estimate the regression model with standard errors clustered by banks. Our final estimation sample is reduced to 30 treatment banks and 132 control banks because of missing values. The arguments for SUTVA remain similar to those presented in Section 3.3.1 and hence, in the interest of brevity, are not repeated here.

When estimating how the launch of credit payments bundled in mobile money affects bank performance, we define treatment and control banks the same as the previous analysis and then estimate the following model. For bank i in country j at time t,

$$EBITDA_{ijt} = \beta_0 + \beta_1 After_t + \beta_2 Loanlaunch_i \cdot After_t + \delta K_{jt} + \tau_y + \tau_q + \alpha_i + \epsilon_{ijt}, \quad (5)$$

where all variables are identical to those in Equation 4, except that $Loanlaunch_i$ stands for the binary variable representing whether the credit payment service is included in the bundle of mobile money for firm i, and

β_2 captures the key effect of interests. The final sample is comprised of 3 treatment banks and 11 control banks. The arguments for SUTVA remain similar to those presented in Section 3.3.1 and hence, for the sake of brevity, have not been repeated here.

3.3.3. End-User Welfare. To assess the impact of mobile money on end-user welfare, we estimate a difference-in-differences regression model (see Equation (6)). The fact that the two surveys (2011 and 2014) consist of different samples of respondents limits us from analyzing the data at the individual level. Furthermore, we are also subject to the following data constraints at the country level. First, the World Bank survey did not include specific questions related to mobile money in the first study in 2011. Second, the survey did not identify the specific mobile money currently used by the respondents in its second study in 2014. To address this issue, we identified markets that had their first mobile money deployed between 2011 and 2014 from the mobile money launch data. We compare the change in the dependent variables between the treatment group (i.e., the countries in which mobile money was launched between 2011 and 2014) and the control group (i.e., the countries in which no mobile money was launched until the end of 2014). We include *GDP* and *Population* in the model to control for country heterogeneity. Our outcome measures (*savings* and *loans*) are on a percentage scale, and we observe that in our sample about 7% of the observations have a value of 0 for the dependent variable. Thus, we specify a Tobit regression model with 0 as the lower bound (Amemiya 1985). In particular,

$$DV_{kt} = \begin{cases} DV^*_{kt}, & if \quad DV^*_{kt} > 0 \\ 0, & if \quad DV^*_{kt} \le 0 \end{cases}, \quad where$$
$$DV^*_{kt} = \beta_0 + \beta_1 Treatment_k + \beta_2 After_t + \beta_3 Treatment_k \cdot After_t + \nu D_{kt} + \epsilon_{kt}, \tag{6}$$

where k indexes the countries (i.e., $Treatment_k = 1$ for countries in which mobile money was launched between 2011 and 2014 and 0 for countries in which no mobile money was launched until 2014), t indexes year (i.e., $After_t = 1$ for 2014 and 0 for 2011), DV represents each of the dependent variables described in Section 3.2.1, D is the vector of control variables described above, and β_3 is the coefficient of interest to be estimated.

4. Results

4.1. Performances of MNOs and Banks and End-User Welfare

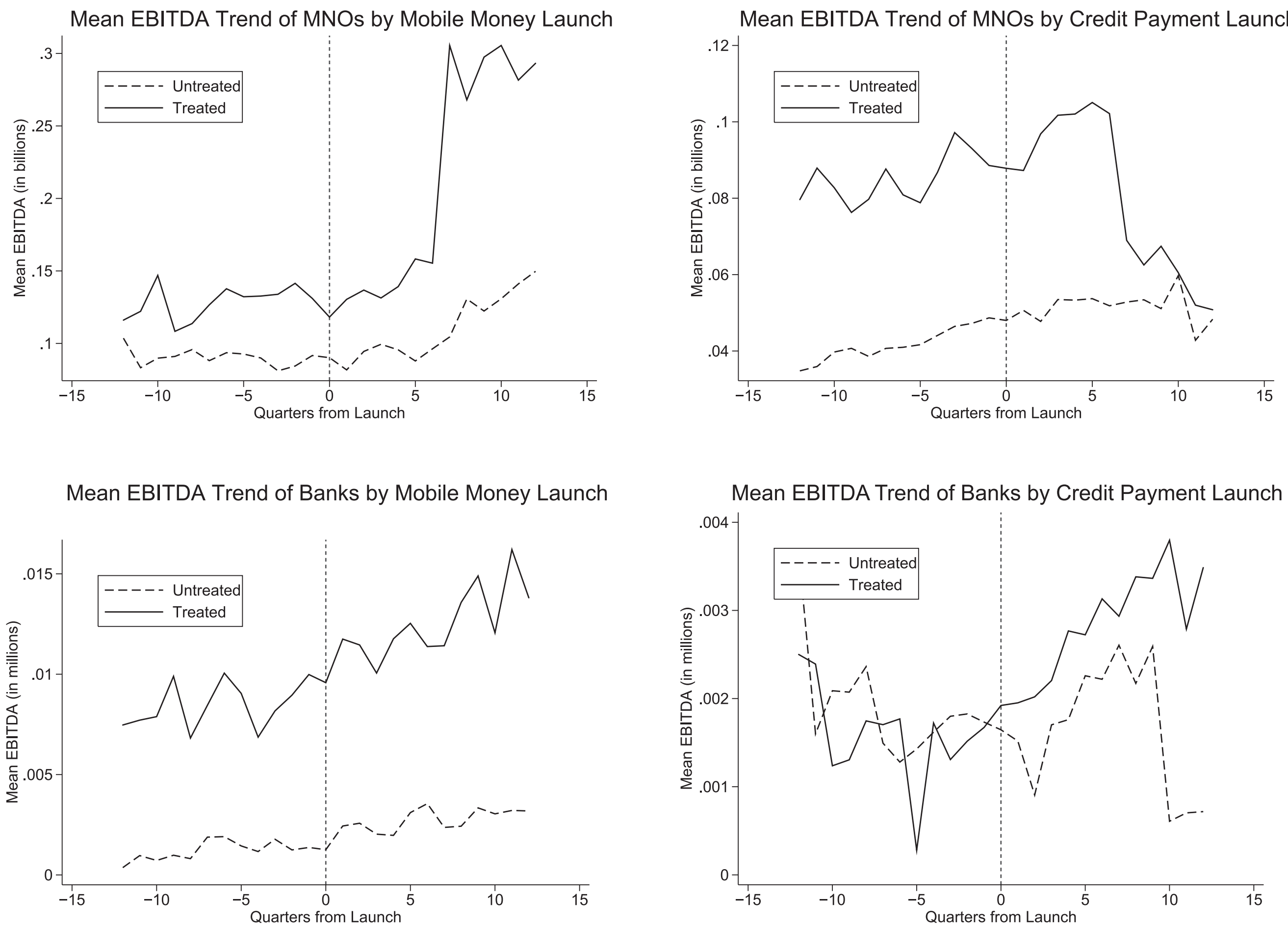
In this section, we first show how the average *EBITDA* changes over time for the treated and control value chain members (i.e., MNOs and banks) before and after the launch of mobile money and credit payment services. In Figure 2, the four plots suggest that the launch of mobile money leads to increased average *EBITDA* for MNOs and banks relative to the matched control groups, whereas the launch of credit payment service leads to increased average *EBITDA* for banks but not for MNOs. Prior to estimating Equations (2)–(5), we also formally tested the parallel trend assumption. We do not find significant differences (at the $p < 0.05$ level) in *EBITDA* between the treated groups and control groups in the pretreatment window (see Web Appendix WA2), a finding that confirms that the parallel trend assumptions are satisfied.

We summarize our main results for MNOs, banks, and end users. First, we report the estimates of Equations (2) and (3) in part A, Table 3. We find that launching mobile money has a positive impact on the *EBITDA* ($\beta = 1.07 \times 10^8$, $p < 0.05$) of MNOs. However, the MNOs that provide the credit payment bundle with mobile money on average have lower *EBITDA* ($\beta = -1.76 \times 10^7$, $p < 0.01$) after the services are launched, compared with MNOs that do not provide this bundle. These results indicate that although mobile money and credit payments can act as complements when offered by one member as bundles (Guiltinan 1987), they may not be complements when multiple members are involved in the value chain.

Second, we report the estimates of Equations (4) and (5) in part B, Table 3. We find that mobile money has a positive impact on the *EBITDA* ($\beta = 46{,}712.99$, $p < 0.05$) of banks. Furthermore, banks that allow end users to make credit payments on average achieve significantly higher *EBITDA* than those that do not ($\beta = 2{,}155.50$, $p < 0.01$). These results suggest that although banks benefit from providing mobile money, offering credit payments as a bundle may create an alternative source of revenue and significantly improve their *EBITDA*. This indicates that bundling services may benefit banks (unlike the MNOs) because of complementarity. These findings contribute to the existing literature of bundling and complementarity of services in the value chain, where the benefits of bundling may not be evenly distributed to benefit all members.

Third, we report the estimates of Equation (6) in Table 4, which extends the analysis of mobile money to end users and provides a holistic view of mobile money and its impact on the value chain. The results show that the launch of mobile money significantly improves the annual savings of the poor and old populations ($\beta = 8.368$, $p < 0.01$ and $\beta = 8.524$, $p < 0.05$, respectively). Moreover, it significantly improves both the annual savings and annual loans of the population with low education ($\beta = 14.333$, $p < 0.01$ and $\beta = 10.273$, $p < 0.05$, respectively). As additional analyses, we use paired sample t-tests to compare the average population loans before and after the launch of mobile money (see Web

Figure 2. Model-Free Evidence



Appendix WA6). The results suggest that with mobile money, people on average receive higher loans ($t = 2.021$, $p < 0.05$). To summarize, the difference-in-differences and t-tests results suggest that mobile money increases end-user welfare by increasing their savings and loans. These results empirically corroborate prior research findings and anecdotal evidence that mobile money improves end-user welfare (Mbiti and Weil 2013, Jack et al. 2013, Jack and Suri 2014).

4.2. Robustness Checks

We demonstrate the robustness of the effects of mobile money on the performance of MNOs by (1) making minor changes to the matching specifications and (2) using alternative methods of instrumental variables regression and a propensity score matching model with estimates of the average treatment effect on the treated MNOs (Heckman et al. 1997, Sianesi 2004).

Our matching results in the main analysis may depend on the matching sequence, which starts the matching from quarter 1 up to quarter 59. Thus, it is possible that the quality of the matches declines in later quarters, which may bias the matched control group. To demonstrate that the results are robust to the matching sequence, we also specify a reversed matching sequence that starts from quarter 59 down to quarter 1. We find consistent effects of launching mobile money ($\beta = 1.63 \times 10^8$, $p < 0.05$) and credit payments ($\beta = -1.69 \times 10^7$, $p < 0.01$) on the *EBITDA* of MNOs. We also test the robustness of results with one nearest neighbor in the propensity score matching and find qualitatively consistent results: $\beta = 5.89 \times 10^7$ ($p < 0.10$) and -1.08×10^7 ($p < 0.05$) for the launching of mobile money and credit payments on *EBITDA* of MNOs, respectively. In addition, we estimated all models without the sampling frequency weights. The estimated effects are qualitatively consistent with our main analyses for mobile money ($\beta = 1.10 \times 10^8$, $p < 0.05$) and credit payments ($\beta = -2.04 \times 10^7$, $p < 0.01$). We also use one nearest-neighbor matching without replacement as a robustness test. Matching without replacement tends to be more biased than matching with replacement when there are significant differences in propensity score distribution between the treated and control groups (Smith and Todd 2005, Caliendo and Kopeinig 2008). We find qualitatively consistent

Table 3. Results for MNO Performance and Bank Performance

	A. MNO performance		B. Bank performance	
	Model (1)	Model (2)	Model (3)	Model (4)
Variables	Dependent variable: EBITDA		Dependent variable: EBITDA	
After	-1.05×10^{8}**	5,601,801	−62018.17***	−1314.117*
	(4.65×10^{7})	(3,765,469)	(17,650.67)	(691.244)
*Mobilemoney*After*	1.07×10^{8}**		46,712.99**	
	(4.80×10^{7})		(18,155.92)	
*Loanlaunch*After*		-1.76×10^{7}***		2155.502***
		(4,333,153)		(711.833)
GDP	0.000*	0.000***	1.04×10^{-7}**	4.29×10^{-9}***
	(0.000)	(0.000)	(4.71×10^{-8})	(1.14×10^{-9})
Population	0.499	0.356	−0.001***	0.000
	(1.542)	(0.568)	(0.000)	(0.000)
Interest	−4,696,741*	704,654.5		
	(2,403,717)	(55,5023.9)		
HHI	−45,847.71*	6,235.016*		
	(24,540.37)	(3,274.659)		
Bconcen			666.996***	−2.478
			(205.564)	(13.485)
Constant	−3,412,080	-1.10×10^{8}		
	(2.05×10^{8})	(1.71×10^{9})		
Year/quarter fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Observations	879	527	5,488	467
Within R^2	0.603	0.791	0.097	0.649
Number of firms	31	17	162	14

Note. Clustered robust standard errors are in parentheses.
*$p < 0.10$; **$p < 0.05$; *** $p < 0.01$.

results for mobile money ($\beta = 3.89 \times 10^7$, $p < 0.10$) and credit payments ($\beta = -1.08 \times 10^7$, $p < 0.05$). Finally, we use alternative sets of covariates to match the MNOs' propensities to launch mobile money and credit payment services. The results, reported in Web Appendix WA3, are qualitatively consistent for mobile money ($\beta = 2.40 \times 10^7$, $p < 0.05$) and credit payments ($\beta = -2.66 \times 10^7$, $p < 0.05$).

The instrumental variables regression allows us to specify an extensive model with all country-level and firm-level variables and tackle multiple endogenous regressors with instrumental variables. However, given our research context and data limitations, we use instrumental variable regressions as a robustness test because the available instrumental variables may not perfectly satisfy the exclusion restriction. In particular, we specify panel data fixed effects models to analyze the data on MNOs. Our dependent variables are the natural logged *EBITDA* (*lnEBITDA*) and revenue (*lnRevenue*). Our focal independent variables are *Mobilemoney* and *Loanlaunch* which are coded as 1 for all quarters after the respective service is launched for the treated group and as 0 otherwise. We believe that certain variables (*Mobilemoney* and *Total connections*) in the model could be endogenous due to unobservable factors (e.g., firm age and reputation); we therefore used instrumental variables to correct for the potential bias from endogeneity. For both dependent variables, we find consistent results for mobile money ($\beta = 0.807$, $p < 0.05$) and credit payments ($\beta = -0.621$, $p < 0.05$). Specifications, instrumental variable (IV) regression test details, and results are reported in the Web Appendix WA4.

To estimate the average treatment effect on the treated, we run a propensity score matching for each quarter, with all firm-level and country-level variables used in our main study and specify 5 nearest-neighbor control MNOs for each treated MNO. At each quarter, we estimate the average treatment effect on the treated (ATT) for MNOs that launch the mobile money or credit payments in the same quarter or earlier. We use *t*-tests, sign tests, and Wilcoxon signed-rank tests to test whether this estimated ATT is significantly different from 0. Web Appendix WA5. The results of this analysis confirm a significant positive impact of mobile money on the performance of MNOs. The propensity score matching model also confirm a significant negative impact of credit payments on MNOs' performance.

Similar to the robustness checks for the analyses on MNOs, we also specify a reversed matching sequence for banks and find consistent effects of launching mobile money ($\beta = 47{,}409.86$, $p < 0.05$) and credit payments ($\beta = 1{,}422.841$, $p < 0.05$) on the *EBITDA* of banks. A model with one nearest-neighbor matching also provides us qualitatively consistent results: $\beta = 4.06 \times 10^4$ ($p < 0.01$) and $\beta = 1{,}866.151$ ($p < 0.01$) for

Table 4. Results for End-User Welfare

Variables	Annual savings			Annual loans		
	Poor 40%	Old	Low education	Poor 40%	Old	Low education
After	12.271*** (2.897)	12.727*** (3.104)	3.706 (2.940)	4.911** (2.331)	4.416** (2.234)	−3.149 (2.985)
Treatment	−11.116*** (3.951)	−10.391** (3.951)	21.458*** (4.400)	2.123 (3.820)	5.914 (3.659)	27.931*** (5.917)
After Treatment*	8.368*** (3.849)	8.524** (4.026)	14.333*** (4.081)	2.372 (3.827)	3.301 (3.738)	10.273** (4.518)
GDPpc	-1.37×10^{-9} (1.30×10^{-7})	9.24×10^{-8} (1.64×10^{-7})	5.99×10^{-7}*** (2.03×10^{-7})	9.81×10^{-8} (8.77×10^{-8})	8.98×10^{-8} (7.71×10^{-8})	4.50×10^{-7}* (2.31×10^{-7})
Population	1.28×10^{-8} (9.06×10^{-9})	2.03×10^{-8}*** (8.88×10^{-9})	3.96×10^{-8}*** (1.33×10^{-8})	3.53×10^{-9} (4.54×10^{-9})	1.51×10^{-9} (3.94×10^{-9})	2.10×10^{-8} (1.36×10^{-8})
Constant	30.047*** (2.776)	37.113*** (2.755)	−3.664 (3.600)	31.369*** (1.904)	30.759*** (1.802)	1.373 (4.744)
Observations	194	194	194	194	194	194
R^2	0.126	0.130	0.269	0.041	0.076	0.210
Number of firms	99	99	99	99	99	99

Notes. Clustered robust standard errors are in parentheses. R^2 is based on the McKelvey/Zavonia R^2 (Veall and Zimmermann 1994).
*$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

the launching of mobile money and credit payments on *EBITDA* of banks, respectively. Estimation results without weights yielded consistent results for mobile money (β = 43,925.65, $p < 0.05$) and credit payments (β = 1,884.992, $p < 0.01$). The one nearest-neighbor matching without replacement provided qualitatively consistent results for mobile money ($\beta = 4.17 \times 10^4$, $p <$ 0.05), despite the concerns discussed earlier and poor matching quality. Finally, we use alternative sets of covariates to match banks' propensities to launch mobile money and credit payment services and find that the results (reported in Web Appendix WA3) are qualitatively consistent for mobile money (β = 34,157, $p < 0.05$) and credit payments (β = 1,628, $p < 0.10$).

We use alternative methods of IV regression and propensity score matching model with estimates of the average treatment effect on the treated (Heckman et al. 1997, Sianesi 2004) to demonstrate the robustness of mobile money or credit payments launch effects on banks' *EBITDA*. The IV regression estimates and average treatment effect on the treated, presented in Web Appendix WA4 and WA5, respectively, generally confirm the key effects and demonstrate the robustness of the results to alternative methods.

To check the robustness of the results of end-user welfare, we use the survey data collected in 2014 and conduct a few descriptive analyses in which we compare end users with mobile money accounts to those without accounts. Ideally, we would like to apply alternative measures of the dependent variable to replicate Equation (3). However, many of the variables measured in the 2014 survey are different from those measured in 2011, which does not allow us to estimate a difference-in-differences model. Therefore, we use t-tests for the robustness check and find that compared with other end users, end users with mobile money accounts have higher agricultural transactions (t = 5.098, $p < 0.01$), receive higher domestic remittances (t = 6.008, $p < 0.01$), and are more likely to raise emergency funds (i.e., funds of one-twentieth of gross national income (GNI) per capita in local currency) within a month (t = 3.904, $p < 0.01$). We provide the detailed results of these t-tests in Web Appendix WA6; the results suggest that mobile money effectively facilitates transactions among end users.

4.3. Time-Varying Effects of Launching Mobile Money

Although the main findings indicate that there are benefits associated with launching mobile money, it is not clear when these benefits are realized. For example, credit payments require time for the customer base to grow, as the unbanked population needs to be familiar with these services. Banks also need time to refine their understanding and credit scoring algorithms to make better approval judgments (Cook and McKay 2015).

Therefore, as an extension, we study the time-varying effects of launching these services to understand the potential benefits of mobile money over time.

To study the time-varying impacts of mobile money on MNOs, we specify the following equation to estimate the average *EBITDA* at each quarter in the postlaunch period. For firm i in country j, at quarter t:

$$EBITDA_{ijt} = \beta_0 + \mathrm{B}\mathrm{T}_t + \Theta Mobilemoney_i \cdot \mathrm{T}_t + \delta Z_{jt} + \tau_y + \tau_q + \alpha_i + \epsilon_{ijt}, \tag{7}$$

where T_t represents a vector of quarterly time dummies that equal 1 for quarters after launching mobile money and equal 0 for quarters before launching the services. Thus, B captures the vector of the average *EBITDA* in the postlaunch quarters, relative to the average *EBITDA* in the prelaunch quarters. The vector Θ captures the time-varying effects of launching mobile money, relative to the control group. The remaining components of Equation (7) are the same as those in Equation (2). We estimate the model via a panel data fixed effects regression with robust standard errors clustered at the firm level. The estimations of the time-varying effects of launching mobile money on *EBITDA* of banks and launching credit payments on *EBITDA* of MNOs and banks follow similar specifications, which are provided in the Web Appendix WA7.

We report the estimates of time-varying effects (for 16 quarters) in Table 5. The effects of launching mobile money on the *EBITDA* of MNOs are significant in almost all 16 quarters. These results suggest that MNOs could expect a positive return in the first year after launching mobile money. The effects of bundling credit payments to mobile money are consistently negative and significant. That is, MNOs that launch credit payments may experience increasing loss of profits over time, relative to MNOs that launch mobile money only. The effects of launching mobile money on the *EBITDA* of banks are mostly positive and stationary. These results suggest that banks should expect a quite steady profit increase after launching mobile money. The effect of launching credit payments on the *EBITDA* of banks is increasing. This result suggests that, relative to banks that only provide mobile money, banks that provide the bundle of credit payments should expect a profit boost.

5. Discussion and Conclusions

Our findings indicate that MNOs have increased *EBITDA* after launching mobile money, but the increase in *EBITDA* is attenuated when mobile money is bundled with credit payments. Launching mobile money has a significant impact on banks' *EBITDA* and launching credit payments further boosts banks' *EBITDA*. This asymmetric distribution of benefits persists over time, and therefore, banks are in an advantageous position to benefit from the synergy between the bundled services.

It is important to acknowledge that mobile money is still a nascent business that policy makers are struggling to regulate. That MNOs have been able to sustain mobile money business under attenuated profits from bundled services may be related to inevitable interdependence between value chain members in the early stages of innovation, as proposed by the resource dependence theory (Pfeffer and Salancik 2003). This interdependency arises from the distinct set of resources that each party holds. MNOs have the expertise in technological platforms and agent network management capabilities, whereas banks have expertise in navigating financial regulations and liquidity management. In addition, our findings indicate that the end users can also benefit from mobile money. Particularly, the poor, old, and less-educated members of the population may have increased income and more access to financial services such as loan services. Such gains by the end users may facilitate the sustainable growth of mobile money in the value chain, of which the end users are also a critical component. Overall, we find that mobile money can be beneficial to the value chain and its members; mobile money also can be bundled with other value chain services, such as credit payments, to take advantage of the potential complementarity. The benefits of the bundling, however, need to be evaluated with more nuance toward individual members in the value chain because complementarity may not exist for every member involved (Mantovani and Ruiz-Aliseda 2016).

Our research contributes to the literature by taking a novel value chain perspective of understanding mobile money innovations in the developing world. The lack of resources, infrastructure, and effectively functioning institutions is often associated with cost-sensitive consumers, uncertain markets and social environments, and a high proportion of the population outside of the formal economy (Barrett et al. 2015). Therefore, such innovation can successfully create value only when it is cost-efficient, flexible under uncertainty, and sufficiently inclusive (Barrett et al. 2015). More importantly, because the innovation is delivered by the value chain, it is sustainable only when it benefits all members of the value chain. Our results indicate that mobile money does directly create value for MNOs, participating banks, and end users in the value chain. However, we find that complementary services such as credit payments could potentially cause asymmetric distribution of the created value between members in the value chain. The bundling of mobile money and credit payments leads to service complementarities, which create value not only for the end users (Chernev 2005, Liu et al. 2010, Xu et al. 2010) but also for some of the members of the value chain,

Table 5. MNO and Bank Performance: Impacts of Mobile Money Services over Time[a]

	MNO				Bank			
	Mobile money		Credit payment service		Mobile money		Credit payment service	
Time since launch	Effect	SE	Effect	SE	Effect	SE	Effect	SE
t+1	$1.09 \times 10^{8***}$	(3.85×10^{7})	-1.03×10^{7}	(8.78×10^{6})	$5.21 \times 10^{4**}$	(2.27×10^{4})	$8.85 \times 10^{2*}$	(4.79×10^{2})
t+2	$9.96 \times 10^{7**}$	(3.85×10^{7})	$-1.44 \times 10^{7*}$	(7.18×10^{6})	$5.49 \times 10^{4*}$	(3.16×10^{4})	$1.26 \times 10^{3***}$	(3.75×10^{2})
t+3	$1.30 \times 10^{8***}$	(4.53×10^{7})	$-1.78 \times 10^{7**}$	(7.04×10^{6})	$4.47 \times 10^{4**}$	(1.90×10^{4})	$1.79 \times 10^{3**}$	(7.33×10^{2})
t+4	$1.19 \times 10^{8***}$	(3.87×10^{7})	$-1.61 \times 10^{7*}$	(8.59×10^{6})	$5.78 \times 10^{4***}$	(2.16×10^{4})	$1.53 \times 10^{3***}$	(2.83×10^{2})
t+5	$1.18 \times 10^{8***}$	(4.06×10^{7})	$-2.37 \times 10^{7***}$	(6.42×10^{6})	3.26×10^{4}	(2.32×10^{4})	$1.74 \times 10^{3***}$	(5.53×10^{2})
t+6	$1.39 \times 10^{8**}$	(5.29×10^{7})	$-2.31 \times 10^{7***}$	(6.55×10^{6})	$4.37 \times 10^{4**}$	(2.14×10^{4})	$1.66 \times 10^{3**}$	(6.49×10^{2})
t+7	$1.45 \times 10^{8***}$	(5.14×10^{7})	$-2.13 \times 10^{7***}$	(6.44×10^{6})	3.63×10^{4}	(2.36×10^{4})	$2.18 \times 10^{3***}$	(4.10×10^{2})
t+8	$1.20 \times 10^{8*}$	(7.05×10^{7})	$-2.10 \times 10^{7***}$	(5.62×10^{6})	$6.23 \times 10^{4***}$	(2.36×10^{4})	$1.18 \times 10^{3**}$	(7.55×10^{2})
t+9	$1.49 \times 10^{8**}$	(6.15×10^{7})	$-3.57 \times 10^{7***}$	(6.08×10^{6})	$6.15 \times 10^{4***}$	(2.11×10^{4})	$2.67 \times 10^{3***}$	(4.56×10^{2})
t+10	$1.55 \times 10^{8**}$	(7.31×10^{7})	$-2.70 \times 10^{7***}$	(5.47×10^{6})	$2.48 \times 10^{4*}$	(1.33×10^{4})	$2.71 \times 10^{3**}$	(7.06×10^{2})
t+11	$1.53 \times 10^{8**}$	(7.24×10^{7})	$-3.45 \times 10^{7***}$	(6.21×10^{6})	$6.07 \times 10^{4**}$	(2.76×10^{4})	$4.15 \times 10^{3***}$	(7.69×10^{2})
t+12	$1.40 \times 10^{8**}$	(6.44×10^{7})	$-3.89 \times 10^{7***}$	(4.39×10^{6})	1.10×10^{4}	(3.46×10^{4})	$3.29 \times 10^{3***}$	(7.02×10^{2})
t+13	$1.45 \times 10^{8**}$	(6.46×10^{7})	$-4.57 \times 10^{7***}$	(6.21×10^{6})	$3.97 \times 10^{4***}$	(1.49×10^{4})	$3.37 \times 10^{3***}$	(6.63×10^{2})
t+14	9.37×10^{7}	(7.22×10^{7})	$-3.82 \times 10^{7***}$	(3.72×10^{6})	$4.43 \times 10^{4***}$	(1.56×10^{4})	$3.77 \times 10^{3***}$	(7.24×10^{2})
t+15	7.95×10^{7}	(5.36×10^{7})	$-4.62 \times 10^{7***}$	(6.69×10^{6})	$4.49 \times 10^{4**}$	(1.84×10^{4})	$4.23 \times 10^{3***}$	(8.38×10^{2})
t+16	6.20×10^{7}	(5.70×10^{7})	$-4.62 \times 10^{7***}$	(4.76×10^{6})	$5.53 \times 10^{4**}$	(2.47×10^{4})	$5.38 \times 10^{2***}$	(8.45×10^{2})

Note. Clustered robust standard errors are in parentheses.
[a]Estimates of the other variables are omitted for brevity but are available on request.
$*p < 0.10$; $**p < 0.05$; $***p < 0.01$.

such as banks (Milgrom and Roberts 1990, Xu et al. 2010). However, the bundling of complementary services does not create added value for all members of the value chain (e.g., MNOs). We contribute to the extant literature on bundling and complementarity by being the first to empirically demonstrate that benefits from value-added bundled services could be asymmetrically distributed between different members of the value chain, and such asymmetric distribution of benefits could persist over time. Future research could identify the exact mechanisms underlying the asymmetric distribution, as well as exploring effective means to resolve this issue.

5.1. Managerial Implications

Our results provide several implications for managers of MNOs and banks as well as policy makers in the developing countries. Most importantly, although the launch of mobile money may create value for MNOs and banks, the value creation mechanisms and processes for them are different. Managers of MNOs should be encouraged to launch mobile money, which will likely yield higher, persistent profits. Banks may also profit from the launch of mobile money, and these benefits may spread to the end users, particularly, the old, the poor, and the less-educated members of the population. Managers of banks should expect additional and persistent gains from the launch of credit payments, but the managers of MNOs should not. However, MNOs earn about 10 times more benefits from launching mobile money service than the loss experienced by being associated with a credit payment service. The banks earn about 5% more profit over the benefits earned due to launching mobile money. This extra benefit increases in the long-term, which provides incentives for banks to continue the bundled services. Therefore, the dependency between MNOs and banks in creating and sharing value from mobile money and bundled services dictates that the MNOs would need to cooperate with banks and vice versa. To lower the persistent asymmetric benefits, bank managers could consider compensation mechanisms such as revenue sharing or profit sharing contracts for the MNOs to adopt the credit payments with mobile money (Ngigi 2012). Policy makers in developing countries should encourage coordination and promote a fair distribution of benefits between these two important members in the value chain to sustain the growth and the continued success of mobile money.

5.2. Limitations and Future Research

Our paper has a few limitations. First, we have only binary data for the main independent variables, that is, the launch of mobile money and credit payments. Future research may study the impact of heterogeneity in these services by using continuous measures. Second, we have data for only a few performance measures because most of the firms in the sample are in developing countries and are not publicly traded. As more data become available, future research may study the impact of launching mobile money and other bundled services on a broader set of performance measures. Third, mobile money has expanded over the years to incorporate various service features such as government-to-person cash disbursement, bulk cash disbursement for businesses, and utility and bill payment services. Given the limitations of our data set, we are unable to assess the impact of these service features. Future research could explore the value contributed by these service features. Fourth, our samples for the difference-in-differences analyses are small, limited by missing values. The results from instrumental variable regressions based on a much larger sample show the robustness of our main results. However, future research is needed to enhance the generalizability of our findings. Fifth, because of limited data on firm behavior, this research offers little insights on establishing the underlying mechanism of why the bundled services create an asymmetric distribution of benefits between value chain members. Further research with rich behavioral data could pursue this path. Finally, we did not have information on the initiator of mobile money in the value chain. Future research conducted with additional information may provide valuable insights on how structure affects the value chain outcomes.

5.3. Conclusions

Our paper makes several contributions to the operations literature and to industry practice. To the best of our knowledge, we are the first to empirically evaluate the impact of mobile money and credit payment services on the value chain and, especially, on the performance of MNOs and banks. Combining the benefits for the end users, this study sheds much-needed insights on the value chain performance of mobile money innovations. In addition, this research provides new and in-depth insights on the bundling of services and complementarity along the value chain. The findings of this paper provide implications for managers of MNOs and banks and for relevant policy makers. We hope that this paper spurs further academic research that examines innovations in value chains of other sectors.

References

Adengo J, Kamukama E (2016) Uganda has highest mobile money fraud cases – report. *Daily Monitor* (September 9). http://www.monitor.co.ug/Business/Uganda-has-highest-mobile-money-fraud-cases–report/688322-3374002-3ugtokz/index.html.

Adner R (2006) Match your innovation strategy to your innovation ecosystem. *Harvard Bus. Rev*. 84(4):98–107.

Ahn JH, Han SP, Lee YS (2006) Customer churn analysis: Churn determinants and mediation effects of partial defection in the Korean mobile telecommunications service industry. *Telecomm. Policy* 30(10):552–568.

Almazan M, Vonthron N (2014) *Mobile Money Profitability: A Digital Ecosystem to Drive Healty Margins* (Groupe Speciale Mobile Association, London).

Amemiya T (1985) *Advanced Econometrics* (Harvard University Press, Boston).

Anderson EG, Parker GG (2013) Integration and cospecialization of emerging complemenetary technologies by startups. *Production Oper. Management* 22(6):1356–1373.

Austin PC (2011) An introduction to propensity score methods for reducing the effects of confounding in observational studies. *Multivariate Behav. Res*. 46(3):399–424.

Bae J, Insead MG (2004) Partner substitutability, alliance network structure, and firm profitability in the telecommunications industry. *Acad. Management J*. 47(6):843–859.

Balasubramanian K, Drake D (2015) Mobile money: The effect of service quality and competition on demand. Working paper No. 15-059, Harvard University, Boston.

Barrett M, Davidson E, Prabhu J, Vargo SL (2015) Service innovation in the digital age: Key contributions and future directions. *Management Inform. Syst. Quart*. 39(1):135–154.

Beck T, Demirgüç-Kunt A, Levine R (2006) Bank concentration, competition, and crises: First results. *J. Banking Finance* 30(5): 1581–1603.

Bhargava HK (2016) Mixed bundling of two independent valued goods. *Management Sci*. 59(9):2170–2185.

Caliendo M, Kopeinig S (2008) Some practical guidance for the implementation of propensity score matching. *J. Econom. Surv*. 22(1):31–72.

Cassiman B, Veugelers R (2016) In search of complementarity in innovation strategy: Internet R&D and external knowledge acquisition. *Management Sci*. 52(1):68–82.

Chernev A (2005) Feature complementarity and assortment in choice. *J. Consumer Res*. 31(4):748–759.

Cook T, McKay C (2015) *How M-Shwari Works: The Story So Far*. Access to Finance FORUM: Reports by CGAP and Its Partners (Consultative Group to Assist the Poor, Washington, DC).

Demirgüç-Kunt A, Klapper LF (2012) Measuring financial inclusion: The global findex database. Working paper, World Bank Policy Research, Washington, DC.

DeYoung R, Lang WW, Nolle DL (2007) How the Internet affects output and performance at community banks. *J. Banking Finance* 31(4):1033–1060.

Evans DS, Heckman JJ (1984) A test for subadditivity of the cost function with an application to the Bell system. *Amer. Econom. Rev*. 74(4):615–623.

Fang E, Palmatier RW, Steenkamp JBEM (2008) Effect of service transition strategies on firm value. *J. Marketing* 72(5):1–14.

Fries S, Taci A (2005) Cost efficiency of banks in transition: Evidence from 289 banks in 15 post-communist countries. *J. Banking Finance* 29(1):55–81.

Gawer A, Cusumano MA (2002) *Platform Leadership: How Intel, Microsoft, and Cisco Drive Industry Innovation* (Harvard University Press, Boston).

Genakos C, Valletti T (2011) Testing the "waterbed" effect in mobile telephony. *J. Eur. Econom. Assoc*. 9(6):1114–1142.

Gerpott TJ, Rams W, Schindler A (2001) Customer retention, loyalty, and satisfaction in the German mobile cellular telecommunications market. *Telecomm. Policy* 25(4):249–269.

Gidvani L, Pasti F, Hodiwalla F, Aneja D, Kayal G (2016) *The Business Case for Payments Banks in India* (Groupe Speciale Mobile Association, London).

Grzybowski L (2005) Regulation of mobile telephony across the European Union: An empirical analysis. *J. Regulatory Econom*. 28(1):47–67.

Guiltinan JP (1987) The price bundling of services: A normative framework. *J. Marketing* 51(2):74–85.

Heckman J, Ichimura H, Todd P (1997) Matching as an econometric estimator: Evidence from evaluating a job training programme. *Rev. Econom. Stud*. 64(4):605–654.

Hernando I, Nieto MJ (2007) Is the Internet delivery channel changing banks' performance? The case of Spanish banks. *J. Banking Finance* 31(4):1083–1099.

Hess AM, Rothaermel FT (2011) When are assets complementary? Star scientists, strategic alliances, and innovation in the pharmaceutical industry. *Strategic Management J*. 32(8):895–909.

Hirano K, Imbens GW (2001) Estimation of causal effects using propensity score weighting: An application to data on right heart catheterization. *Health Services Outcomes Res. Methodol.* 2(3): 259–278.

Hirano K, Imbens GW, Ridder G (2003) Efficient estimation of average treatment effects using the estimated propensity score. *Econometrica* 71(4):1161–1189.

Hughes N, Lonie S (2007) M-PESA: Mobile money for the "unbanked" turning cellphones into 24-hour tellers in Kenya. *Innov. (Camb, Mass).* 2(1–2):63–81.

Imbens GW, Rubin DB (2015) *Causal Inference in Statistics, Social, and Biomedical Sciences* (Cambridge University Press, Cambridge, UK).

Jack W, Ray A, Suri T (2013) Transaction networks: Evidence from mobile money in Kenya. *Amer. Econom. Rev.* 103(3):356–361.

Jack W, Suri T (2014) Risk sharing and transactions costs: Evidence from Kenya's mobile money revolution. *Amer. Econom. Rev.* 104(1):183–223.

James C (1995) When do banks take equity in debt restructurings? *Rev. Finance Stud.* 8(4):1209–1234.

Kumar PR, Ravi V (2007) Bankruptcy prediction in banks and firms via statistical and intelligent techniques—A review. *Eur. J. Oper. Res.* 180(1):1–28.

Lal R, Sachdev I (2015) Mobile money services: Design and development for financial inclusion. Working paper, Harvard University, Boston.

Lee H, Schmidt G (2016) Using value chains to enhance innovation. *Production Oper. Management* 26(4):617–632.

Lee H, Tang CS (2017) Socially and environmentally responsible value chain innovations: New operations management research opportunities. *Management Sci.* 64(3):983–996.

Leishman P (2010) *Is There Really Any Money in Mobile Money?* (Groupe Speciale Mobile Association, London).

Liu H, Chintagunta PK, Zhu T (2010) Complementarities and the demand for home broadband internet services. *Marketing Sci.* 29(4):701–720.

Lonie S, Wagener I (2012) Transformational banking from transformational relationships: Case study of a bank and mobile network operator partnership. *J. Payments Strategic Syst.* 6(4):333–344.

Majumdar SK (1998) On the utilization of resources: Perspectives from the US telecommunications industry. *Strategic Management J.* 19(9):809–831.

Mantovani A, Ruiz-Aliseda F (2016) Equilibrium innovation ecosystems: The dark side of collaborating with complementors. *Management Sci.* 62(2):534–549.

Markides CC, Williamson PJ (1994) Related diversification, core competences and corporate performance. *Strategic Management J.* 15(52):149–165.

Mbiti I, Weil DN (2013) The home economics of e-money: Velocity, cash management, and discount rates of M-pesa users. *Amer. Econom. Rev.* 103(3):369–374.

McGrath F, Lonie S (2013) *Platforms for Successful Mobile Money Services* (Groupe Speciale Mobile Association, London).

Milgrom P, Roberts J (1990) The economics of modern manufacturing: Technology, strategy, and organization. *Amer. Econom. Rev.* 80(3):511–528.

Ngigi G (2012) M-Kesho growth stalls over hitch on profit sharing. *Business Daily* (March 25), http://www.businessdailyafrica.com/M-Kesho-growth-stalls-over-hitch-on-profit-sharing–/539552-1373474-g6ndp1z/index.html.

Pfeffer J, Salancik GR (2003) *The External Control of Organizations: A Resource Dependence Perspective* (Stanford University Press, Palo Alto, CA).

Prasad A, Venkatesh R, Mahajan V (2010) Optimal bundling of technological products with network externality. *Management Sci.* 56(12):2224–2236.

Robins JM, Rotnitzky A (1995) Semiparametric efficiency in multivariate regression models with missing data. *J. Amer. Statist. Assoc.* 90(429):122–129.

Rosenbaum PR, Rubin DB (1983) The central role of the propensity score in observational studies for causal effects. *Biometrika* 70(1): 41–55.

Rosenbaum PR, Rubin DB (1985) Constructing a control group using multivariate matched sampling methods that incorporate the propensity score. *Amer. Statist.* 39(1):33–38.

Rubin DB (2001) Using propensity scores to help design observational studies: Application to the tobacco litigation. *Health Services Outcomes Res. Methodol.* 2(3):169–188.

Sealey C, Lindley J (1977) Inputs, outputs, and a theory of production and cost at depository financial institutions. *J. Finance* 32(4): 1251–1266.

Sharpe SA (1997) The effect of consumer switching costs on prices: A theory and its application to the bank deposit market. *Rev. Indust. Organ.* 12(1):79–94.

Sianesi B (2004) An evaluation of the Swedish System of active labor market programs in the 1990s. *Rev. Econom. Statist.* 86(1):133–155.

Smith JA, Todd PE (2005) Does matching overcome LaLonde's critique of nonexperimental estimators? *J. Econometrics* 125(1):305–353.

Stremersch S, Tellis GJ (2002) Strategic bundling of products and prices: A new synthesis for marketing. *J. Marketing* 66(1):55–72.

Stuart EA (2010) Matching methods for causal inference: A review and a look forward. *Statist. Sci.* 25(1):1–21.

Stuart EA, Huskamp HA, Duckworth K, Simmons J, Song Z, Chernew M, Barry CL (2014) Using propensity scores in difference-in-differences models to estimate the effects of a policy change. *Health Services Outcomes Res. Methodol.* 14(4):166–182.

Veall MR, Zimmermann KF (1994) Practitioners corner: Goodness of fit measures in the tobit model. *Oxf. Bull. Econom. Statist.* 56(4):485–499.

Velayuthan D (2010) Coopetition between MNO's and banks in provisioning mobile money services today." *Africa Telecoms* (Issue 8), http://www.euroinvestor.dk/pdf/cse/11369052_21811.pdf.

Winship C, Radbill L (1994) Sampling weights and regression analysis. *Sociol. Methods Res.* 23(2):230–257.

Wooldridge JM (2010) *Econometric Analysis of Cross Section and Panel Data* (MIT Press, Cambridge, MA).

Xu X, Venkatesh V, Tam KY, Hong SJ (2010) Model of migration and use of platforms: Role of hierarchy, current generation, and complementarities in consumer settings. *Management Sci.* 56(8): 1304–1323.